

ASEN 5264 - DMU

Homework #4

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Problem 1:

- a) After a large number of pulls, it is reasonable to assume that $\rho_a \rightarrow \theta_a$. However, since we are using an ϵ -greedy policy, we still choose randomly with probability ϵ . With probability $1 - \epsilon$ we will choose greedily, always picking arm 3. With $\epsilon = 0.15$, we can solve for the probability of pulling arm a .

$$\begin{aligned} P(a_1) &= \epsilon/3 = 0.05 \\ P(a_2) &= \epsilon/3 = 0.05 \\ P(a_3) &= \epsilon/3 + 1 - \epsilon = 0.9 \end{aligned}$$

The expected payoff per pull is given as $\mathbb{E}[R]$, where $\mathbb{E}[R]$ is given by:

$$\mathbb{E}[R] = \sum_{k=1}^3 P(R_i)P(a_i) = 0.2 * 0.05 + 0.3 * 0.05 + 0.7 * 0.9 = \boxed{0.655}$$

- b) Again, after a large number of pulls, we have $\rho_a \rightarrow \theta_a$. A softmax policy tells us $P(a_3) \propto e^{\lambda \rho_3} = e^{5\theta_3}$. In order to obtain a valid probability, we must multiply by a normalizing constant $\frac{1}{c}$ where $c = \sum_{k=1}^3 e^{5\theta_k}$.

$$P(a_3) = \frac{e^{5\theta_3}}{\sum_{k=1}^3 e^{5\theta_k}} = \frac{33.11}{40.32} = \boxed{0.821}$$

- c) Starting with a prior distribution $\text{Beta}(1, 1)$, we update the distribution for each arm with $\text{Beta}(1 + w, 1 + l)$ where w and l are the total number of wins and losses respectively. For each arm we have the following beliefs.

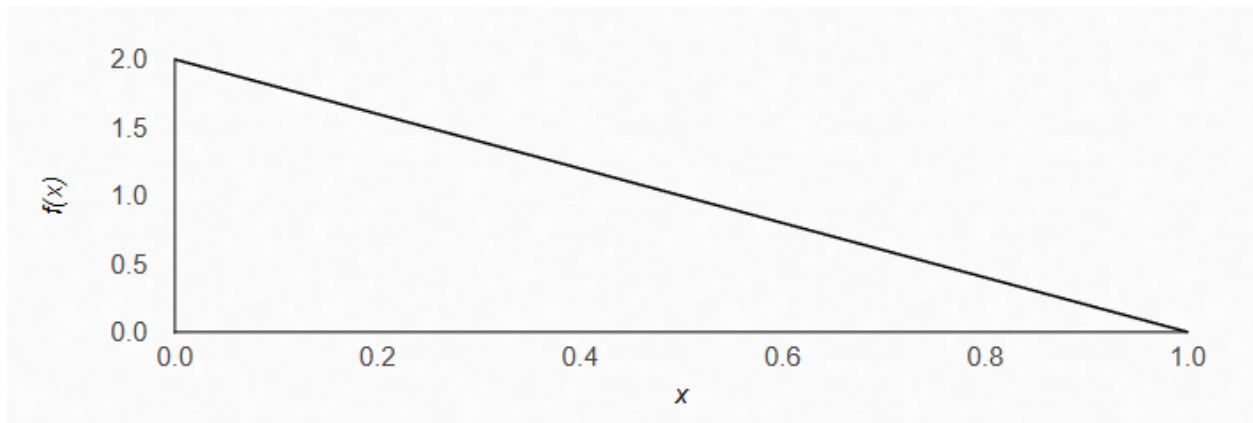


Figure 1: Belief for arm 1

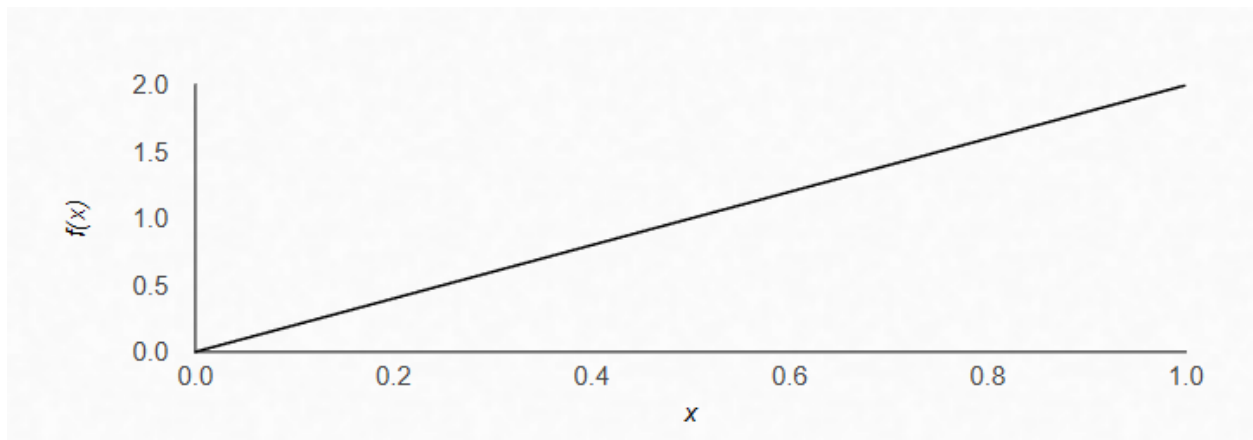


Figure 2: Belief for arm 2

- d) First, we would sample a value $\hat{\theta}_a$ from each belief distribution, then we would choose the arm which had the largest $\hat{\theta}_a$, that is $\arg \max_a \hat{\theta}_a$. Some theoretical values that could be sampled are $\hat{\theta}_1 = 0.1$, $\hat{\theta}_2 = 0.8$, $\hat{\theta}_3 = 0.65$. In this case, Thompson sampling would choose arm 2, observe the outcome and update the belief for arm 2.

Problem 2:

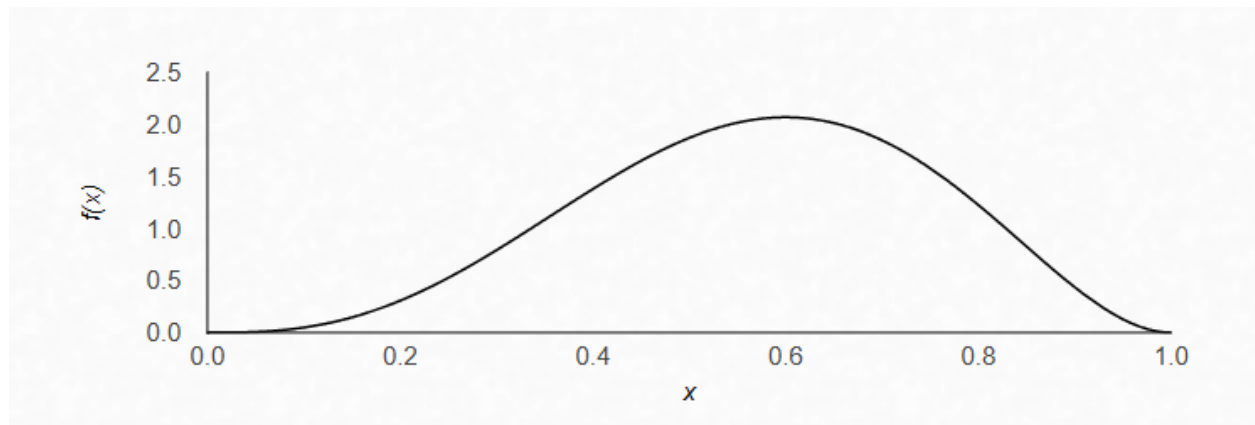


Figure 3: Belief for arm 3